

Functional Cross-Model KV State Handoff: An Evidence-Gated Study of Large-to-Small Transfer

Anonymous Authors

Abstract

Can a smaller language model continue from a runtime key-value (KV) state produced by a larger model without re-prefilling the original context? We study this question as *cross-model state handoff* and separate three claims that are often conflated: engineering replay correctness, representation compatibility, and downstream capability transfer. On one real-GPU Qwen3-4B→Qwen3-1.7B run, external replay of the student’s own KV state is exact on four samples. A ridge map trained on 64 prompts and evaluated on 16 disjoint prompts obtains key cosine 0.9661 and key $R^2 = 0.9282$, but value cosine is only 0.5772 and value $R^2 = -0.9449$. These measurements support an asymmetric linear-compatibility result; they do not establish task retention, capability gain, latency savings, or generalization. We identify two evaluation errors in the original pipeline: global cropping reduced the nominal 512/1024-token experiment to 70 tokens, and a value diagnostic incorrectly used values as both keys and values. We provide corrected variable-length evaluation, standard $\text{softmax}(QK^\top/\sqrt{d})V$ diagnostics, and evidence gates that prevent synthetic scores or proxy timing from entering scientific verdicts. Based on these findings, we propose the most viable next path: learn a functional compatibility base first, then train a bounded advantage residual against real held-out task loss. The present evidence supports continuing the research program, but not yet claiming runtime capability transfer.

1 Introduction

Transformer inference materializes a KV cache during prefill and reuses it during autoregressive decoding [8]. The cache is a major serving object: systems optimize its allocation [4], quantization [7], token selection [6], and reuse within a model [10]. A harder question is whether a runtime state produced by one model can be consumed by another.

We consider a large teacher T , a smaller student S , a long context X , and a short continuation query q . The teacher has already produced $C_T(X)$, while the student is not allowed to read X again. A handoff method constructs

$$\hat{C}_S(X) = F_\theta(C_T(X)), \quad (1)$$

injects it into the student, and evaluates the student’s continuation. This setting is attractive for mid-session model downgrade, repeated reuse, and heterogeneous serving. It is also easy to evaluate incorrectly: high KV cosine need not imply equivalent logits, equivalent logits need not imply task quality, and a CUDA microbenchmark need not imply end-to-end savings.

This paper makes three contributions. First, it introduces an evidence ladder for cross-model handoff and treats missing evidence as an inconclusive result, not as a failed or successful transfer. Second, it reports a real-GPU large-to-small mapping result and diagnoses the K/V asymmetry and two protocol errors that materially affect interpretation. Third, it presents a staged experimental design centered on functional compatibility followed by a real task-trained advantage residual. Our conclusion is deliberately narrow: cross-model linear state compatibility is plausible for keys in the tested pair, while runtime capability transfer remains unmeasured.

2 Related Work and Positioning

The closest concurrent work is no longer only generic KV-cache systems. Heo et al. study cross-model KV transfer within model families using closed-form per-head ridge maps, layer selection, and de-RoPE, and

report receiver task retention and prefill speedups across multiple pairs [3]. Cache-to-Cache (C2C) learns a neural projection, fusion, and gating mechanism for semantic communication between different LLMs [2]. Mixture-of-Translators extends cache translation to heterogeneous models with multiple translation experts and target-side correction [5]. These studies directly establish that cross-model cache mapping is an active and feasible research direction.

Our scope is different and narrower. We do not claim to originate cross-model KV transfer, ridge translation, or de-RoPE mapping. Instead, this paper audits a preliminary large-to-small replication/extension and asks what its artifacts can actually support. Its contribution is the separation of representation recovery, injected task retention, capability gain, and end-to-end systems benefit; a corrected K/V functional diagnostic; and an artifact-level gate that prevents lower-grade evidence from silently satisfying a higher-grade claim. A future competitive evaluation must compare directly with the ridge protocol of Heo et al., C2C, and a heterogeneous translator such as MoT under the same zero-prefill receiver constraint.

3 Problem Formulation and Evidence Levels

3.1 Three distinct targets

We distinguish the following targets.

Replay correctness. The student’s native self-prefill cache is exported, reconstructed, and replayed. Matching logits and tokens test cache capture and injection, but do not involve cross-model mapping.

Compatibility or replacement. A mapped teacher state should approximate the behavior obtained from student self-prefill. The primary metric must be downstream task retention,

$$R_{\text{task}} = \frac{\text{Score}(S \mid \widehat{C}_S(X), q)}{\text{Score}(S \mid C_S(X), q)}. \quad (2)$$

KV cosine, R^2 , CKA, and attention-output cosine are diagnostics, not substitutes for R_{task} .

Capability transfer. The stronger claim requires the handed-off student to outperform student self-prefill on a preregistered held-out task:

$$\text{CHG} = \text{Score}(S \mid \widehat{C}_S) - \text{Score}(S \mid C_S), \quad (3)$$

$$\text{TGRR} = \frac{\text{Score}(S \mid \widehat{C}_S) - \text{Score}(S \mid C_S)}{\text{Score}(T \mid C_T) - \text{Score}(S \mid C_S)}. \quad (4)$$

Positive CHG with a confidence interval excluding zero is the primary scientific endpoint. Positive TGRR indicates recovery of part of the teacher–student gap.

3.2 Evidence gate

Every artifact is assigned one of four roles: measured task evidence, measured representation evidence, proxy evidence, or synthetic/placeholder evidence. A claim is eligible only if all required inputs have the appropriate role.

4 Method: Functional Compatibility Before Advantage

The most viable path separates a stable compatibility base from a later advantage-carrying residual.

Table 1: Evidence required for each conclusion. A lower-grade artifact may diagnose a mechanism but cannot satisfy a higher-grade claim.

Claim	Required evidence	Insufficient substitute
Replay correctness	real logits/tokens	tensor shape only
State compatibility	real injected task score	KV cosine/ R^2
Capability transfer	real held-out CHG, TGRR	simulated scores
System benefit	end-to-end handoff timing	CUDA proxy/formula
Generalization	unique measured model pairs	repeated runs of one pair

4.1 Stage I: compatibility base

For student layer ℓ and head h , selected teacher layers are mixed and mapped separately for keys and values:

$$Z_{\ell h}^K = \sum_{i \in \mathcal{A}(\ell)} w_{\ell i h}^K K_{i h}^T, \quad \hat{K}_{\ell h}^S = Z_{\ell h}^K W_{\ell h}^K, \quad (5)$$

$$Z_{\ell h}^V = \sum_{i \in \mathcal{A}(\ell)} w_{\ell i h}^V V_{i h}^T, \quad \hat{V}_{\ell h}^S = Z_{\ell h}^V W_{\ell h}^V. \quad (6)$$

Keys are de-rotated before mapping and re-rotated in the student coordinate system. Values are not de-rotated. Ridge regression is a useful calibration baseline, but the training target should progress from raw KV reconstruction to functional behavior:

$$\mathcal{L}_{\text{compat}} = \lambda_{\text{task}} \mathcal{L}_{\text{task}} + \lambda_{\text{KL}} D_{\text{KL}}(p_{\text{self}} \| p_{\text{handoff}}) + \lambda_{\text{att}} \|A_{\text{self}} - A_{\text{handoff}}\|_2^2 + \lambda_{\text{KV}} \mathcal{L}_{\text{KV}}. \quad (7)$$

The task and logit terms are primary; the KV term regularizes but does not define success.

4.2 Correct functional diagnostic

Given student queries Q^S , the diagnostic must use a causal and padding-aware mask M , with disallowed entries set to $-\infty$:

$$A(Q, K, V; M) = \text{softmax}(QK^\top / \sqrt{d} + M)V. \quad (8)$$

We report three interventions: K-only $A(Q, \hat{K}, V^S; M)$, V-only $A(Q, K^S, \hat{V}; M)$, and joint $A(Q, \hat{K}, \hat{V}; M)$. The corrected implementation applies the causal mask by default and accepts an explicit key-validity mask. Random queries are explicitly labeled proxy; the final experiment records real student queries and their native attention masks.

4.3 Stage II: bounded advantage residual

Only after the base reaches stable real task retention do we train an advantage residual:

$$C_S^* = C_{\text{base}} + \alpha_{\text{max}} \tanh(a) \odot R_{\text{adv}}(C_T). \quad (9)$$

R_{adv} uses separate low-rank K/V branches and RMS calibration. Unlike the preliminary implementation, a must be updated by real task loss; an all-zero, never-trained α is not evidence for or against advantage transfer. We recommend causal layer/head interventions: ablate or inject one candidate subspace at a time and measure paired changes in CHG. This turns “advantage” from a name into a testable, task-causal object.

5 Real-GPU Experimental Evidence

5.1 Setup

The tested pair is Qwen3-4B→Qwen3-1.7B [9]. Both expose eight KV heads with head dimension 128; the teacher and student have 36 and 28 layers, respectively. Models are loaded sequentially on one GPU. The

Table 2: Real measurements from the archived run. “Combined” is the mean of K and V representation metrics, not task retention.

Artifact/channel	Cosine	R^2	Valid interpretation
Student self-KV replay ($n = 4$)	1.0000	–	injection correctness
Mapped K (16 held-out prompts)	0.9661	0.9282	strong linear recoverability
Mapped V (16 held-out prompts)	0.5772	-0.9449	poor ridge reconstruction
Combined K/V diagnostic	0.7716	-0.0083	representation diagnostic only

real mapping run uses 64 calibration prompts and 16 disjoint evaluation prompts. K and V maps are fitted independently with ridge coefficient 10^{-3} . The archived run requested 512/1024-token contexts but the old evaluator cropped every sample to the global minimum, 70 tokens. We therefore treat the result as a short-context pilot and require a corrected rerun before any long-context claim.

5.2 Results and valid interpretation

The key result is encouraging: a simple per-head linear map recovers much of the student key representation on disjoint prompts. The value result is the dominant bottleneck. Negative R^2 means the fitted value map is worse than a mean predictor under this metric, even though its cosine is positive. We do not interpret 0.5772 as a mathematical ceiling. In particular, near-zero raw cross-model cosine does not preclude a learned rotation or a nonlinear functional map.

The archived attention-output values are excluded from the scientific table. The old code evaluated each channel through a two-argument helper that implicitly set $V = K$; consequently the “V attention” branch computed $A(Q, V, V)$ rather than $A(Q, K, V; M)$. Corrected code now reports K-only, V-only, and joint causal interventions and marks deterministic random Q as proxy evidence.

5.3 What the run does not establish

The run does not measure real injected downstream task retention. The reported “retention” was KV cosine divided by self-cosine, whose denominator is one by construction. Synthetic T09 scores therefore cannot establish positive CHG or TGR; CUDA proxy timing cannot establish positive PSR_A ; placeholder geometry cannot explain the failure; and two executions of the same Qwen pair cannot establish two-pair generalization. Under the evidence gate, the current verdict is **inconclusive evidence**, not runtime capability transfer and not a definitive stop result.

6 Revised Experimental Design

6.1 Phase A: make compatibility functional

Use 100–500 calibration prompts and at least 500 held-out task examples. Preserve each prompt’s natural length up to the requested context cap; aggregate ridge sufficient statistics across variable-length samples. Evaluate 512, 1024, 4096, and 8192-token buckets separately. The student self-prefill path is the reference; the handoff path must have zero original-context prefill.

The minimum method set is student self-prefill, teacher inference, text handoff, ridge K/V, a small nonlinear or low-rank base, and selective recompute. For every method report task score, task retention, next-token KL, top- k overlap, and judge/ranking consistency where applicable. Freeze layer mapping, ridge coefficients, ranks, and all thresholds on train/validation.

6.2 Phase B: localize the value failure

Run four controlled interventions: native K/native V, mapped K/native V, native K/mapped V, and mapped K/mapped V. Record real queries and per-layer attention outputs.

Sweep value objectives in this order:

1. ridge reconstruction;
2. low-rank plus bias and scale calibration;
3. attention-output distillation with native K;
4. next-token KL and held-out task loss;
5. a small token/layer recomputation budget.

If K-only remains stable but V-only fails, allocate capacity to V rather than increasing a shared mapper. If failures concentrate in a few layers, selective recompute is a stronger systems path than forcing a universal map.

6.3 Phase C: test advantage only after Gate 1

Require real task retention ≥ 0.90 on held-out data before training the advantage residual. Train on a dataset-level teacher-advantage task selected before test evaluation; never filter test examples by teacher success. Use at least three seeds and paired bootstrap or permutation inference on per-example CHG. The capability gate requires: CHG > 0 , a 95% confidence interval whose lower bound is above zero, TGR > 0 , stable KL/JCR, and replication on a second unique model pair.

6.4 Phase D: measure systems claims end to end

Time complete paths with synchronization, warmup, and repeated trials. For natural handoff, report

$$\text{PSR}_A = 1 - \frac{T_{\text{map}} + T_{\text{load}} + T_q}{T_{\text{prefill},S}(X)}. \quad (10)$$

If the teacher runs only for transfer, include teacher prefill. For repeated reuse, report the measured break-even query count. Include CPU offload, host-to-device transfer, cache conversion, injection, failures, and any fallback recomputation. CUDA proxy kernels remain engineering diagnostics.

7 Discussion

Most viable research path. The data argue against immediately adding a complex “advantage” module. The highest-probability path is a functional compatibility base with independent K/V capacity and selective recomputation, followed by a small task-trained residual. This path is useful under all outcomes. If compatibility and CHG both succeed, it supports runtime capability transfer. If compatibility succeeds but CHG ≤ 0 , it yields a clean boundary: state compatibility does not imply capability compatibility. If values remain incompatible, the work becomes a mechanistic study of cross-model state geometry and the minimum recomputation needed for handoff.

Why the negative value result is valuable. The K/V asymmetry narrows the design space. It suggests that capacity should not be allocated uniformly and that raw reconstruction is not the only useful target. It also motivates layerwise interventions, because value tensors carry the content aggregated by attention and may be more model-specific than keys in this pair.

Relation to KV-cache systems work. Existing systems demonstrate that KV state is a first-class memory and reuse object [4, 6, 7, 10]. Our question is orthogonal: whether a state can cross a model boundary while remaining functionally consumable. Efficient attention kernels such as FlashAttention [1] reduce attention I/O but do not solve semantic cross-model compatibility.

Direct cross-model cache transfer is addressed by concurrent ridge, cache-communication, and translator approaches [2, 3, 5]. Our revised experiment should therefore be read as a rigorously gated replication and boundary study, not as an isolated first demonstration. The key open comparison is whether a functional objective and selective recomputation improve the specific weak-V regime observed here while preserving the zero-prefill receiver contract.

8 Limitations and Reproducibility Status

Only one model pair has real mapping measurements. Replay correctness uses four samples; the mapped evaluation uses 16 held-out prompts; the effective sequence length is only 70 tokens due to the superseded global-cropping protocol. Real task injection scores, real student queries, end-to-end timing, real geometry, multi-turn stability, and a second model pair are absent. The reported values are therefore preliminary engineering and representation evidence. The revised code records per-sample effective lengths, separates metric semantics, blocks non-measured verdicts, rejects real T08 execution until task supervision is connected, loads real geometry only from explicit hidden-state artifacts, and groups generalization runs by model identity and revision without discarding repeats. For real-score execution, the revised pipeline accepts only audited held-out artifacts with exact sample IDs, verified zero receiver prefill, and verified task scoring. System evidence likewise requires synchronized end-to-end handoff records with warmup and at least ten repeats. Generalization aggregates all runs for each model ID plus revision, requires at least three unique seeds, and conservatively fails a pair if any included measured run fails. These paths make the missing experiments executable, but no such new real artifacts are included in the present result table.

9 Conclusion

The current evidence supports continued investigation, but not the claim that runtime capability has already been transferred. Exact self-KV replay shows that the injection mechanism can work. Strong held-out key recovery shows that part of the large-to-small state relation is linearly compatible. Weak value recovery, a short effective context, and the absence of real downstream and end-to-end measurements leave the central hypothesis unresolved. The most feasible next experiment is a functional, independently parameterized K/V compatibility base evaluated by real task behavior, with selective recompute as a safety valve; only then should a bounded, task-trained advantage residual be tested. This evidence-gated formulation makes both positive and negative outcomes scientifically useful.

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A Preregistered Decision Rules

Gate 0 passes only when external replay matches native self-prefill logits and tokens within numerical tolerance. Gate 1 is based on real task retention: ≥ 0.90 passes, $0.80-0.90$ is conditional, and < 0.80 fails. A representation-only run is inconclusive regardless of cosine. Gate 2 requires positive real held-out CHG and TGR, a paired 95% interval above zero, and stable behavior. A systems claim additionally requires end-to-end timing. Generalization requires at least two unique model-and-revision identities, at least three unique seeds per pair, and aggregation of every included run; it is not satisfied by two run IDs or by selecting the best repeat.

B Required Artifact Schema

Each task artifact stores `offline_demo`, `evidence_grade`, metric semantics, model identifiers, data split identifiers, sample counts, effective sequence-length statistics, and timing provenance. The final decision artifact lists all blockers rather than substituting zero for missing measurements. This schema is intended to make accidental overclaiming visible in both automated reports and paper tables.

The task-score artifact additionally records the held-out split, exact sample IDs, method, seed, score, decision, and explicit `zero_prefill_verified` and `task_scoring_verified` flags. The timing artifact records synchronized per-context, per-seed teacher prefill, student prefill, map, load, and query times together with warmup and repeat counts.